

Reviewer 1

Comments

Reviewer 1 - Initial submission, 02/08/2026

Comments for Author

The manuscript presents a hybrid CNN–BiLSTM Attention framework for cross-dataset DDoS detection in IoT networks with Explainable AI. The topic is timely and relevant, and the integration of CNN, BiLSTM, attention mechanisms, and XAI techniques is interesting. The manuscript is generally well organized, and the experiments demonstrate promising detection performance across multiple public IoT datasets. However, several issues should be addressed before the manuscript is suitable for publication.

Major Comments

Although the proposed model reports excellent accuracy, the manuscript should provide stronger statistical validation (e.g., repeated runs with standard deviation, confidence intervals, or statistical significance tests) to demonstrate the robustness of the reported results.

The cross-dataset evaluation is one of the claimed contributions. However, the experimental protocol should be explained more clearly, including the exact training/testing combinations, dataset splits, and whether any data leakage was prevented during feature harmonization.

The computational complexity of the proposed CNN–BiLSTM-attention architecture should be discussed in greater detail. Although latency and energy proxies are reported, comparisons with existing lightweight state-of-the-art methods would strengthen the practical contribution.

The manuscript would benefit from a detailed ablation study showing the individual contributions of the CNN branch, BiLSTM branch, attention mechanism, and XAI module.

The reproducibility of the work should be improved by providing complete implementation details, including hyperparameter settings, random seeds, hardware configuration, software versions, and training time.

The explainability section should include a more comprehensive discussion of how SHAP, LIME, and ANOVA complement one another and how these explanations can assist cybersecurity analysts in real-world deployments.

The manuscript primarily evaluates offline datasets. The limitations regarding deployment in real-time operational IoT environments should be discussed more explicitly.

Minor Comments

Carefully proofread the manuscript to eliminate minor grammatical and typographical errors.

Some figures contain relatively small text that may reduce readability in the published version. Increasing the font size would improve clarity.

Several recent references from 2025–2026 are included; however, the literature review could better compare the proposed approach against the latest transformer-based and foundation-model-based intrusion detection methods.

Please ensure consistent formatting of equations, symbols, abbreviations, and references

throughout the manuscript.

Include a short discussion on future work, particularly addressing adversarial robustness, continual learning, and deployment on resource-constrained edge devices.

Overall, the proposed work is technically sound and addresses an important problem. After addressing the above comments, the manuscript will be significantly strengthened.

Reviewer 2

Comments

Reviewer 2 - Initial submission, 18/08/2026

See the attached file.

Attached file: MEADIG-D-26-00146_Reviewer_Report.pdf